

HAT-P-54b

R-1891

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_250920 · created 2026-08-03T12:54:43+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460938.9347 +/- 0.0057 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.188 +/- 0.045
Transit depth (Rp/Rs)^2	3.55 +/- 1.71 [%]
Orbital Inclination (inc)	89.93 +/- 1.93
Airmass coefficient 1 (a1)	1.026 +/- 0.031
Airmass coefficient 2 (a2)	-0.02 +/- 0.015
Scatter in the residuals of the lightcurve fit is	3.07 %
Best Comparison Star	#3 - [47, 329]
Optimal Method	PSF photometry
Transit Duration (day)	0.0733 +/- 0.0044

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.188 ± 0.045 vs 0.1572 (deviation 0.62σ)
Duration fit vs published	0.0733 d vs 0.075 d (deviation 0.35σ)
Mid-time O-C	13.6 min
Depth SNR	3.8
Residual scatter	3.07 %

Light curve

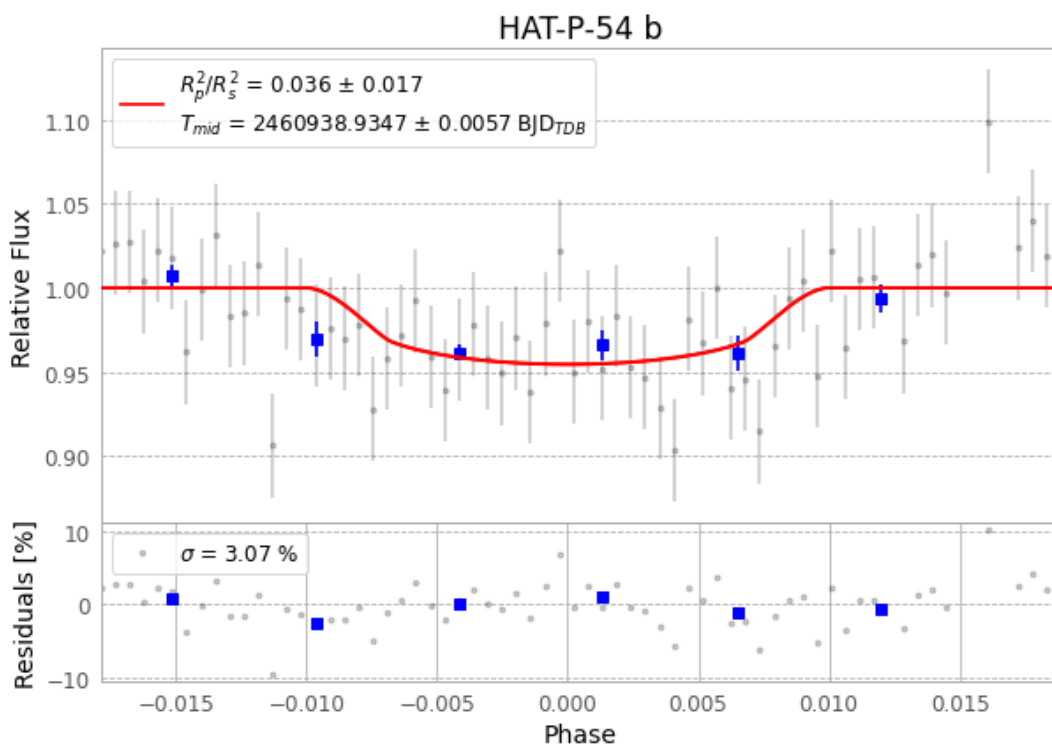


Figure 1 — FinalLightCurve_HAT-P-54 b_2025-09-20.png (SHA-256 e9dc7e44b3520d98...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [322, 201], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b51a08f941f76814...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

68 FITS frames (45 MB), HATP-54250920084815.FITS → HATP-54250920120915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-250920.log (14d146dfb99e...), FinalLightCurve_HAT-P-54 b_2025-09-20.png (e9dc7e44b352...), FinalLightCurve_HAT-P-54 b_2025-09-20.csv (3d01aac130c3...)

Compute resources

Wall time 175.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 12:54Z.